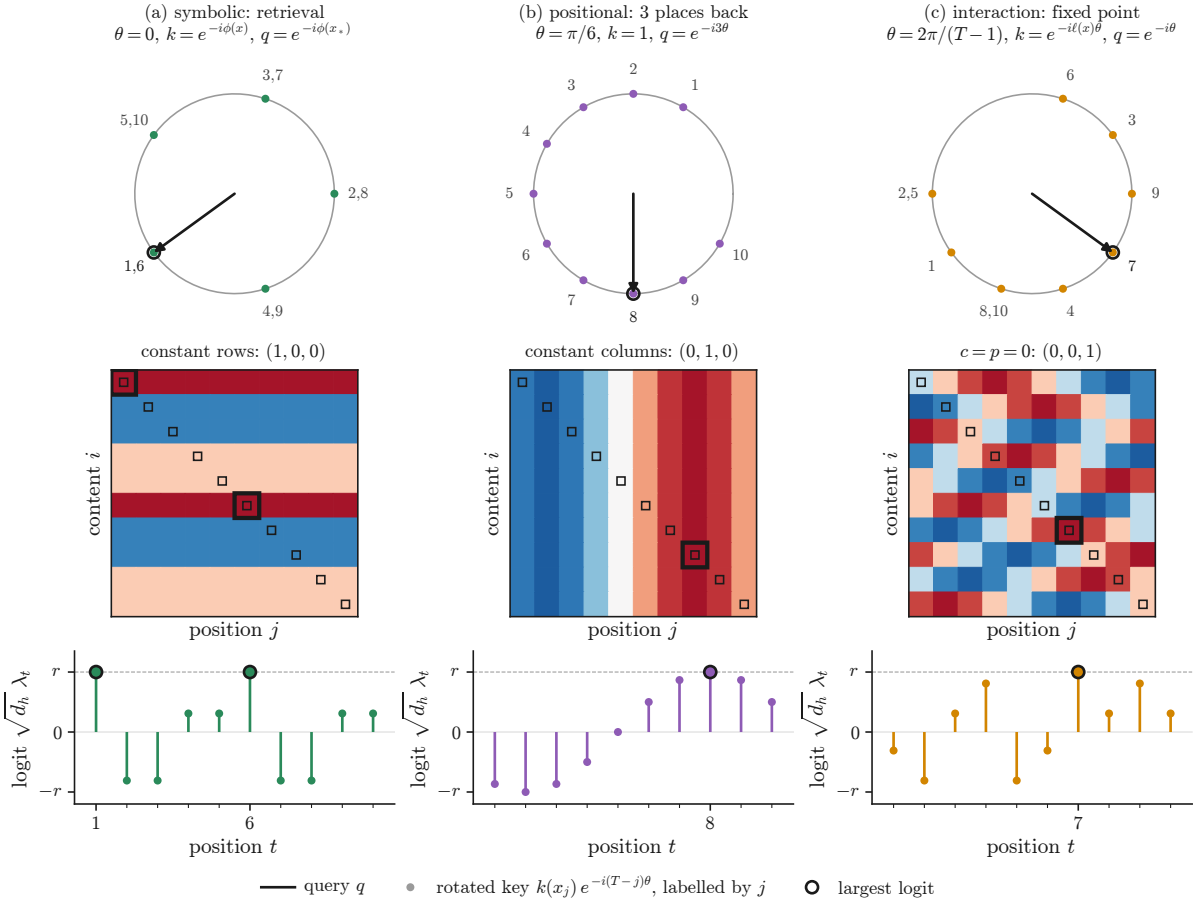
(a) symbolic: retrieval
$\theta = 0$, $k = e^{-i\phi(x)}$, $q = e^{-i\phi(x_*)}$
3,7
5,10
2,8
1,6
4,9
(b) positional: 3 places back
$\theta = \pi/6$, $k = 1$, $q = e^{-i3\theta}$
2
3
1
4
5
6
10
7
9
8
(c) interaction: fixed point
$\theta = 2\pi/(T-1)$, $k = e^{-i\ell(x)\theta}$, $q = e^{-i\theta}$
6
3
2,5
9
1
7
8,10
4
constant rows: (1, 0, 0)
constant columns: (0, 1, 0)
$c = p = 0$: (0, 0, 1)
content i
position j
logit $\sqrt{d_h}\,\lambda_t$
r
0
$-r$
1
6
8
7
position t
query q
rotated key $k(x_j)\,e^{-i(T-j)\theta}$, labelled by j
largest logit